A Gated Feature Fusion Network With Meta-Encoder and Self-Calibrating Cross Module for Building Change Detection in Remote Sensing Images

Jinsheng Deng, Gufeng Gong, Guoxiong Zhou, Miying Yan, and Liujun Li

***Abstract*— Remote sensing (RS) technology plays a critical role in monitoring our constantly changing world, with building change detection (BCD) being a pivotal application that contributes to urban planning, disaster management, and environmental monitoring. In tasks of BCD in high-resolution RS images, it is faced with challenges such as complex backgrounds, redundancy in feature fusion information, and imbalance of positive and negative samples. Utilizing high-resolution RS images for BCD tasks remains challenging. Therefore, a new BCD network named Meta-SGNet with Siamese architecture is proposed. First, a self-calibrating cross module (SCCM) algorithm is proposed to extract the morphological characteristics of buildings in RS images effectively. Subsequently, the gated feature fusion module (GFFM) is proposed to fuse the features of bitemporal buildings dynamically. Finally, a self-learning meta-encoder (Meta-E) is proposed, which uses a meta-learning algorithm to guide the encoder to encode the bitemporal RS image to better pay attention to the learning of positive samples of building changes to improve the accuracy of BCD. Experimental results show that Meta-SGNet outperforms ten state-of-the-art (SOTA) BCD methods on three datasets (Google-CD, WHU-CD, and LEVIR-CD). In the practical application, we acquired 40 pairs of high-resolution image pairs via Google Earth API for a real BCD task. The application results show that Meta-SGNet can accurately capture the range of building changes and shows high adaptability and the ability to quickly detect building changes in different scenarios.**

***Index Terms*— Building change detection (BCD), gated feature fusion model (GFFM), meta-encoder (Meta-E), meta-learning, self-calibrating cross module (SCCM).**

I. Introduction

THE changes in land cover types resulting from urban development can profoundly impact Earth's resources and the ecological environment [1]. Land cover change detection is a crucial component of RS applications, providing essential information for tasks such as updating fundamental geospatial data [2], urban planning [3], environmental impact assessments [4], and emergency disaster response decision-making [5]. Buildings are a major type of urban land surface. During urbanization, buildings are continuously constructed, reconstructed, and demolished. Relying on manual field investigations of building changes is time-consuming and lacks accuracy, failing to meet the needs of modern city management [6]. Therefore, intelligent interpretation of building changes is still an important research content for the application of remote sensing (RS) technology [7]. RS technology has become increasingly mature, and the ability to observe the ground has reached an unprecedented level [8]. Currently, RS data sources are more diverse, and the information extraction technology is developing rapidly. However, the existing studies have shown that the increased resolution of RS images has not significantly increased the interpretation accuracy [9], [10], [11], [12]. As resolution improves, noise and background information interference become more evident. The phenomena of "the same object with different spectra" and "the same spectra with different objects" still exist [13]. Using high-resolution RS imagery to detect building changes remains a challenge, requiring further in-depth research.

In recent decades, many scholars have dedicated themselves to research on change detection in RS imagery utilizing deep learning techniques [14]. This includes the application of methods such as convolutional autoencoders (CAEs) [15], stacked autoencoders (SAEs) [16], deep belief networks (DBNs) [17], pixel-based deep neural networks (DNNs) [18], recurrent neural networks (RNNs) [19], convolutional neural networks (CNNs) [20], and fully convolutional networks (FCNs) [21] to RS image change detection, achieving good results. Additionally, deep learning technology exhibits high robustness to data noise [22], [23]. The model can autonomously learn most parameters with minimal manual intervention [24], [25], thereby achieving superior building

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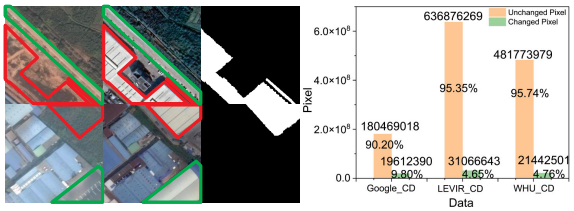


Fig. 1. (a) Image of a bitemporal building, with before the change on the left, after the change in the center, and a labeled image on the right. The red boxes represent the changed areas and the green boxes represent the areas that are easily misdetected. (b) Statistical graph of the number of changed and unchanged pixels.

change detection (BCD) across various RS image datasets. Increasingly, experts and scholars concur that deep learning-based change detection methods can partially automate these tasks [26]. Despite the efficiency and accuracy of deep learning methods in BCD tasks, numerous challenges remain.

A. RS Images Have Complex Backgrounds

RS images typically include diverse features and landscapes such as buildings, vegetation, roads, and water bodies. Many of these are complex backgrounds unrelated to building changes. As shown in Fig. 1(a), different classes of features and landscapes may overlap together, making the information in the image blurred. This can lead to difficulties for traditional neural networks to accurately extract the features of buildings, thus reducing the accuracy of the detection task, which adds to the difficulty of application in BCD tasks.

B. Information Redundancy in Bitemporal Feature Fusion

In the BCD task, bitemporal features need to be fused. Normally, there is overlap and similarity between features extracted at two different times. Traditional feature fusion methods simply stack these feature values together, failing to filter unchanged information effectively, which prevents the model from fully utilizing the important differences of features at different time points.

C. Positive and Negative Sample Imbalance

In the BCD tasks, there is usually an imbalance between positive and negative samples. The sample sizes of the positive samples (i.e., the changing part of the building) and the negative samples (i.e., the unchanged component other than the building change) in the data are quite different, as shown in Fig. 1(b); we count the number of changing pixels and invariant pixels on three datasets (Google-CD [27], LEVIR-CD [28], WHU-CD [29]). As can be seen in Fig. 1(b), the percentage of pixel changes is very low. Because of the limited number of positive samples, traditional deep learning methods may be biased toward predicting negative examples. This bias occurs because predicting most samples as negative increases overall accuracy given the current data distribution. Consequently, the model may fail to accurately capture changes, thereby affecting detection accuracy.

To address the problem of complex backgrounds [30], Chen et al. introduced the spatiotemporal attention (STA) structure in STA-Net [28] by utilizing spatiotemporal dependence in combination with self-attention. The STA structure computes attentional weights between any two pixels across different times and locations. This process generates more discriminative features, effectively capturing the spectrotemporal relationships within the input bitemporal RS images. However, this strategy does not completely exploit the feature maps' local and global information, such as building edges. Zheng et al. [31] proposed the HFAB module by merging the spatial attention (SA) and high-frequency augmentation (HF) modules in their proposed HFA-Net. The SA could guide the model to pay attention to the buildings and highlight the high-frequency portion of the buildings, i.e., the boundaries, through the HF that improves the feature extraction capability. However, although the module is able to clearly recognize objects at the boundaries, it may also misclassify surface coverings with similar spectral and morphological features to buildings as buildings. Ye et al. [32] innovatively used 3-D convolution to treat dual-phase images as time frames. This method allows the network to extract and fuse features in dual-phase images at the same time, which greatly improves the effect and detection accuracy of feature extraction [32]. Luo et al. [33] proposed a differential fusion attention (DFA) that can effectively capture the subtle features in multitemporal hyperspectral images by extracting temporal features, and improve the accuracy of change detection.

To solve the problem of information redundancy during bitemporal feature fusion, Xu et al. [34] proposed a feature enhancement module (FEM) for feature fusion in their FERA-Net by combining the local attention module (LAM) and the global attention module (GAM). This structure obtains the initial feature fusion information by adding and subtracting the dual time-phase features and then aggregates the local and global information by LAM and GAM, respectively. Experimental results show that the algorithm has good results. However, some valuable details or feature information may be lost when adding and subtracting features from the bitemporal features, resulting in loss of information. Wang et al. [35] proposed a feature interleaved fusion module (FIFM) that can effectively fuse the shared features and difference features of dual-temporal images to generate more discriminative fusion features and improve the accuracy of change detection. Luo et al. [36] proposed a multiscale attention fusion (MSAF) module that can adaptively fuse multiscale features, which further improves the accuracy of change detection. Zhang et al. [37] propose SwinSUNet, a pure transformer-only U-shaped network for BCD. SwinSUNet uses the Fusion module to fuse bitemporal features. The Fusion module mainly consists of a connectivity layer (connecting the two feature maps), a linear projection layer (to reduce the dimensionality), and two Swin Transformer blocks (for effective fusion). This structure enhances the traditional concatenation method by adding two layers of the Swin Transformer to filter the features. This approach has yielded excellent results in BCD tasks, improving the integration of features from different time periods.

To solve the positive and negative sample imbalance problem, some existing methods aim to enhance the model's

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effectiveness by synthesizing CD samples, e.g., CDNet + IAug [38], and transferring a pretrained model, e.g., SaDL [39], which is actually adding change pixels to make it more robust. In contrast, Han et al. [40] introduced progressive foreground-balanced sampling (PFBS) techniques that do not incorporate change information. Oubara et al. introduced a channel attention mechanism to focus on the features of the changing region and ignore irrelevant information [41]. These methods aid the model in precisely capturing the features of the foreground image during the initial stages of the training process. However, focusing on foreground samples may lead to overfitting [42]. If the number of foreground samples is very small, the model may overly focus on these samples. This can lead to good performance on the training data but poor generalization on actual test data.

To address the above problems, we propose a novel BCD Meta-SGNet with Siamese structure based on FCN to recognize building change objects effectively. The main contributions of this work can be summarized as follows.

1) A self-calibrating cross module (SCCM) is proposed, which first generates more discriminative representations by adaptively establishing dependencies between the remote space and the channels surrounding each spatial location, and then effectively aggregates the remote contextual information by coordinating the relationship between the horizontal and vertical weight distributions on the spatial dimensions, thus reducing the likelihood of architectural change detection errors due to the complex context.
2) A gated feature fusion module (GFFM) is designed. This module enhances the model's learning of task-relevant features and suppresses the learning of irrelevant features by calculating the gating value when performing feature fusion. Avoiding the interference of information redundancy improves the detection accuracy of the model and makes it better adapted to the BCD task.
3) For the first time, a meta-learning algorithm is combined with an encoder and proposes a meta-encoder (Meta-E) that can learn autonomously. The meta-learning algorithm guides the encoder to autonomously learn how to mine feature information from RS images, focuses on the learning of change categories to better cope with the change situation of a few categories, and improves the accuracy of BCD.

II. Methods

A. Meta-SGNet

Since the FCN-based Siamese network performs great in handling the change detection task, the Meta-SGNet with a typical encoder–decoder backbone was designed by basing this framework. In the network input part, images obtained at different time ($T1$ and $T2$) are convolved using 1×1 convolution kernels to increase the number of channels. Then, these images are then sent to encoders in the same architecture, which share weights. The encoder is constructed from a feature extraction block and a downsampling block, and these two operations extract the features of the image. Before the first downsampling, to be able to capture the local features and edge information of the original input image, we use a module consisting mainly of 3×3 convolutional layers for feature extraction, which is crucial for the subsequent feature enhancement and downsampling. Before the second, third, and fourth downsampling stages, the semantic building information is guided by the insertion of the SCCM. This module integrates building features to focus on the changing characteristics of the buildings. Prior to each downsampling step, we save the feature maps for each layer. After the fourth downsampling, the process enters the decoding phase. Each layer's feature mapping from $T1$ to $T2$ was fused by using GFFM. Then, skip connections are made across the fusion layer feature mapping so that the bottom layer feature maps and the top layer feature mapping can be utilized. These skip connections let CNN to fully consider different levels of features during the decoding process, which found useful in U-Net [43], [44]. In addition, feature mapping within various layers typically have varying spatial dimensions due to sampling layer operations such as pooling, which is able to recover targets with various spatial dimensions. The output layer mainly consists of three convolutional layers (kernel sizes of 1×1 and 3×3). After the decoding is completed, the feature maps are restored to the same size as the input image, and the output layer generates the final prediction. The overall network parameters and network architecture are shown in Fig. 2(a) and (b).

1) Self-Calibrated Cross Module (SCCM): Aiming at the problem of complex and diverse backgrounds in RS images, we propose an SCCM for feature extraction. First, the input information is filtered through self-calibrated convolution (SCC) [45] to filter the complex background and perform feature extraction on the building. Although the SCC module establishes the relationship between remote spatial and channel dependencies around each spatial location, it struggles to aggregate contextual information over long distances. To address this, we designed a cross-attention module (CAM) to coordinate the horizontal and vertical weight distribution in the spatial dimension. This module effectively aggregates long-distance context information, expanding the influence of important features while reducing the impact of irrelevant features. The SCCM is shown in Fig. 2(c).

a) Self-calibrated convolution: Ordinary convolution uses the same convolutional kernel parameters at each location [46], resulting in a fixed receptive field that can only perceive the input image at a local scale. This design limits the ability of ordinary convolution to capture global or larger scale contextual information. To address this limitation, we employ self-calibrating convolution, which efficiently captures contextual information at each spatial location. The benefits of self-calibration are mostly in three areas. First, in comparison to the traditional convolution, the self-calibration operation employs the calibration operation in (4), which allows each spatial location to adaptively use the surrounding information context as an embedding of the potential space and as a scalar in the original scale spatial response, as well as modeling the interchannel dependencies. This effectively expands the field of view of the self-calibrating convolutional layer and enhances the perceptual capability. Second, the self-calibration operation does not incorporate global context in information collection; instead, it focuses

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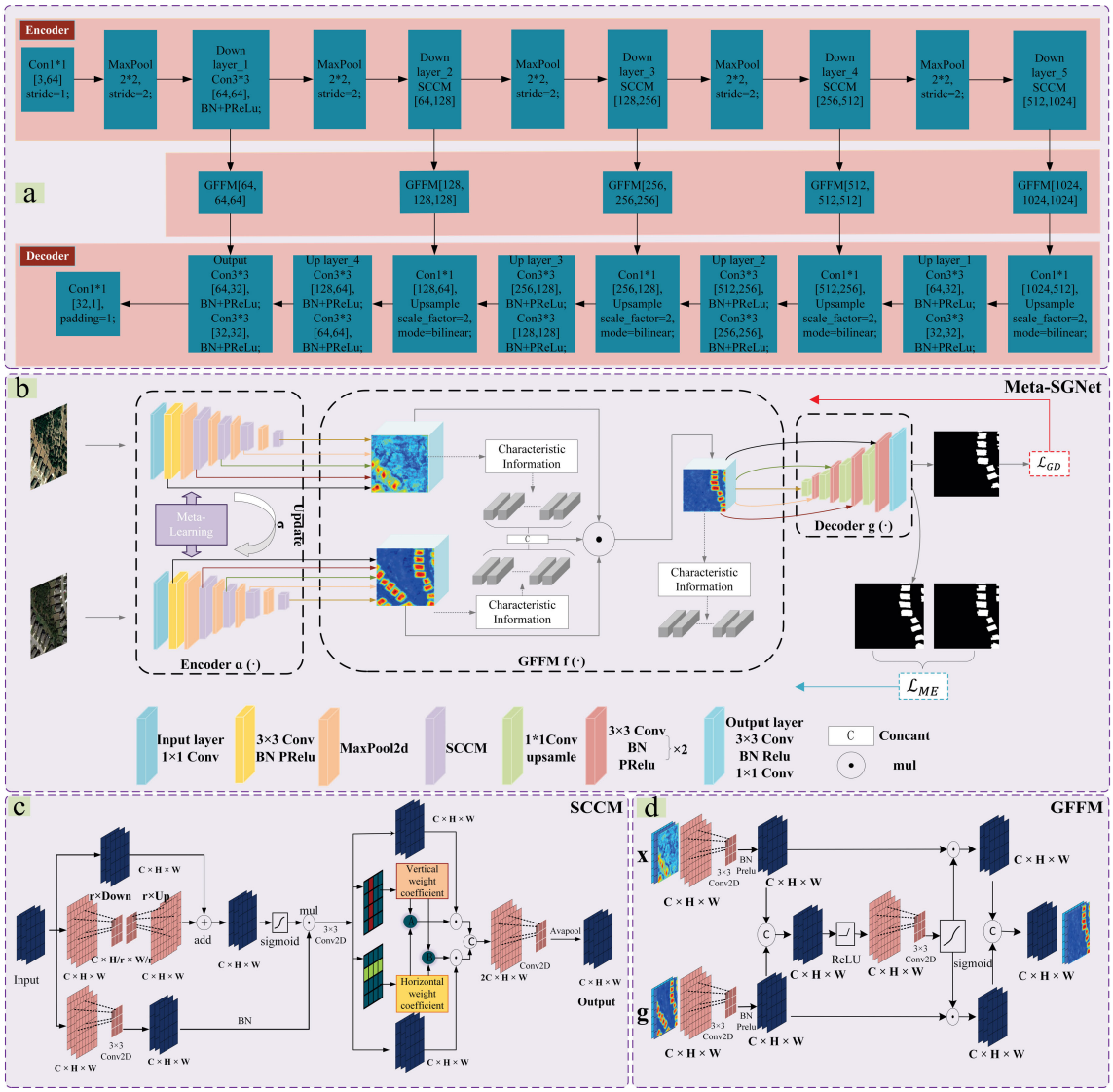


Fig. 2. Structure diagram of Meta-SGNet. (a) Network parameters. (b) Network structure. (c) SCCM. (d) GFFM.

on the context around each spatial location, thereby avoiding contamination from irrelevant regions to some extent. Third, the method encodes multiscale information by fusing data from two different scale spaces through a self-calibration operation. This is important in related tasks such as image segmentation. It allows the model to better understand the features at different scales, thus improving the performance of feature extraction. The SCC calculation process is as follows, where K_1, K_2, and K_3 denote different convolutional kernels.

Given the input X, it is first processed using average pooling with a window size of $r \times r$ and a step size of r. The formula is shown below

$$T = \text{Avg Pool}_r(X) \tag{1}$$

the feature transformation on T is based on K_1

$$X' = \text{Up}(T * K_1) \tag{2}$$

where $\text{Up}(\cdot)$ is a bilinear interpolation operator that maps the downsampled small-scale space to the original feature space. The calibration operation can be expressed as follows:

$$Y' = F_2(X) \cdot \sigma(X + X') \tag{3}$$

where $F_2(X) = X * K_2$, σ is the sigmoid function, and "$\cdot$" denotes cell multiplication. As (3) shows, we use X' as residuals to form the weights used for calibration and found this to be beneficial. The final output after calibration can be expressed as follows:

$$Y = F_3(Y') = Y' * K_3. \tag{4}$$

b) Cross-attention module: In addition to aggregating contextual information around each spatial location, we obtain horizontal and vertical weight distribution relations in the spatial dimension using cross focusing. c_I is the vertical weight. c_{II} is the horizontal weight. We multiply the two weights to get the weight multiplication strategy A, and take the larger of the c_I and c_{II} to get the maximum weight strategy B. The feature weights A and B are multiplied with the input image features to get the weighted image features, and splicing

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and fusion are performed. The computational expression is as follows:

$$c_i = \sum_{j=1}^{n} \frac{\exp(e_{(i,j)})}{\sum_{k=1}^{n} \exp(e_{(ik)})} h_j \tag{5}$$

$$A = \mathrm{mul}(c_I * c_{II}) \tag{6}$$

$$B = \max(c_I, c_{II}) \tag{7}$$

$$\mathrm{CA} = \mathrm{concatenate}([\mathrm{input} * A], [\mathrm{input} * B]) \tag{8}$$

where $e_{(i,j)}$ denotes the weight coefficient assigned by the attention mechanism, and pixel j denotes the sequence of features. i denotes the features at a given moment in time, and h_j represents the hidden layer information of the feature sequence j. c_I denotes the weight coefficient of the vertical attention mechanism in the feature sequence ($c_I = \{c1, c2, \ldots, ci-1, ci\}$), and c_{II} denotes the weight coefficient of the horizontal attention mechanism in the feature sequence ($c_{II} = \{c1, c2, \ldots, ci-1, ci\}$). mul denotes the multiplication operation of the weight coefficients. max denotes the maximum operation of the weight coefficients.

The SCCM is capable of adaptively building dependencies between remote spaces and channels around each spatial location. It merges richer information to generate more discriminative representations and harmonizes the relationship between horizontal and vertical weight distributions in the spatial dimension. This efficient aggregation of long-distance contextual information reduces the likelihood of segmentation errors. Its effectiveness experiments are shown in Section III-F1.

2) Gated Feature Fusion Module: In tasks such as BCD, the feature mapping of each layer of $T1$ and $T2$ images extracted by the encoder needs to be fused, and feature fusion is an advanced feature integration method, showing excellent performance in solving complex data challenges. However, traditional feature splicing methods, such as simple concatenation, although capable of fusing features in some cases, suffer from a core problem: information redundancy [47]. This issue limits their application in complex tasks. Information redundancy is a major concern with traditional concatenation methods. Simple splicing of different features directly is likely to lead to the information of certain features being introduced repeatedly, which triggers the model to pay excessive attention to the repeated information and reduces the generalization ability of the model. In addition, when there is some correlation between different features, the splicing method may not be able to learn appropriate weight assignments from them, leading to further exacerbation of the feature redundancy problem. Therefore, we propose a GFFM that can dynamically inhibit the model from learning task-irrelevant parts and enhance the learning of task-relevant features through gating values when performing feature fusion, thus finely integrating the features and avoiding the plague of information redundancy, which can help to boost the model's performance and make it better adapted to tasks such as complex BCD.

GFFM is shown in Fig. 2(d). First, we divide the feature images x and g extracted from the two-phase image into two inputs, map the features to a unified feature space through linear transformation, and then connect the mapped features and pass through a ReLU activation function to fuse the two features in the feature dimension. Then, the sigmoid is adopted to normalize the attention coefficient to calculate the gated value of the feature, and the weighted feature is obtained by applying the gated value to the initial feature map. Finally, these weighted features are spliced together, and the final fusion features are obtained after further processing. The calculation expression is as follows:

$$q^l = \psi\left(\sigma_1\left(W_x x_i^l + W_g g_i^l + b\right)\right) + b_\psi \tag{9}$$

$$\alpha_i^l = \sigma_2\left(q^l\left(x_i^l, g_i^l; \theta\right)\right) \tag{10}$$

$$\mathrm{DA} = \mathrm{concatenate}\left(\left[x_i^l * \alpha_i^l\right], \left[g_i^l * \alpha_i^l\right]\right) \tag{11}$$

where $\sigma_1(x_i) = \max(0, x_i$ corresponds to the ReLU activation function, and $\sigma_2(x_i) = (1/(1 + \exp(-x_i)))$ corresponds to the sigmoid activation function. A group of parameters θ characterizes the gating value which contain the following: linear transformations $W_x \in (F_{\mathrm{int}} * F_{\mathrm{int}})$, $W_g \in (F_g * F_{\mathrm{int}})$, $\psi \in (R^{2F_{\mathrm{int}} * F_{\mathrm{int}}})$, and the bias term $b_\psi \in R, b \in R(F_{\mathrm{int}})$. The linear transformation is computed by performing a channel-by-channel 1×1 convolution of the input tensor, where the connected features x^l and g^l are linearly mapped to the $R^{(F_{\mathrm{int}})}$-dimensional intermediate space. $\alpha_i \in [0, 1]$ denotes the attention coefficients. These attention coefficients are used to identify salient image regions and prune the feature responses to retain only task-specific relevant activations. l represents the lth layer.

GFFM calculates the gated value α_i to better deal with the problem of information redundancy that occurs during feature fusion, and its effectiveness experiments are shown in Section III-F2.

3) Self-Learning Meta-E: In BCD tasks, unchanged regions usually make up the vast majority [48]. This leads to the model being more likely to learn the unchanged features during training, while ignoring the few changed regions. For this problem, traditional encoders may not be able to extract fine features adequately because these categories appear less frequently in the training data, making it difficult for the model to learn an effective feature representation for them and affecting the model's ability to generalize on new data. In this regard, to improve fine feature extraction and enhance the generalizability, we use a meta-learning [49] approach to guide the encoder in encoding images. Meta-learning is a method to learn how to learn. By simulating multiple tasks in the training stage, it finds the optimal parameters and appropriately initializes the encoder to guide the encoder to mine feature information from the graph to better cope with the changes in a few categories.

As shown in Fig. 2(b), we propose an encoder structure that can learn autonomously, i.e., Meta-E. We need to realize the BCD through an encoder–decoder. For this purpose, we construct an encoder Encode and train it by a meta-learning approach. In training, we first fix the Encode $a_\sigma(\cdot)$, and train GFFM $g_\phi(\cdot)$ and Decode $f_\varphi(\cdot)$ by backpropagating L_{GD}, where σ, ϕ, and φ denote the network parameters of $a_\sigma(\cdot)$, $g_\phi(\cdot)$, and $f_\varphi(\cdot)$, respectively. L_{GD} represents binary cross-entropy loss with logits (BCEWithLogitsLoss()). Then, $g_\phi(\cdot)$ and $f_\varphi(\cdot)$ are fixed, and $a_\sigma(\cdot)$ is trained by computing its gradients with respect to the performance of $g_\phi(\cdot)$ and $f_\varphi(\cdot)$, and the

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meta-learning updating objective is as follows:

$$\arg_{\sigma} \min(L_{\text{Meta-E}}(f_{\dot{\varphi}}(g_{\dot{\phi}}(a_{\sigma}(T1, T2))), \text{target})) \quad (12)$$

where $f_{\dot{\varphi}}(g_{\dot{\phi}}(a_{\sigma}(T1, T2)))$ denotes the BCD result and target denotes the BCD target. a_{σ}(T1,T2) denotes the feature extraction for $T1$ and $T2$ moment images. $\dot{\varphi}$ and $\dot{\phi}$ denote the parameters corresponding to the GFFM and decoder, respectively, and we update φ and ϕ using the following equation:

$$\dot{\varphi} = \varphi - l \bigtriangledown_{\varphi} (L_{\text{GD}}(f_{\phi}(g_{\varphi}(a_{\dot{\sigma}}(T1, T2))), \text{target})) \quad (13)$$

$$\dot{\phi} = \phi - l \bigtriangledown_{\phi} (L_{\text{GD}}(f_{\phi}(g_{\varphi}(a_{\dot{\sigma}}(T1, T2))), \text{target})) \quad (14)$$

where l is the learning rate shared by φ and ϕ, and $\dot{\sigma}$ denotes the corresponding parameter of the encoder. The idea behind the meta-update objective is to perform the second-order derivative trick to train $a_{\sigma}(\cdot)$. Specifically, we update σ using the second derivative of $\{\varphi, \phi\}$ (i.e., the derivative of the Hessian matrix). We calculate the derivative relative to σ by using the computational graph of the reserved $\{\varphi, \phi\}$. Then, update σ in the following ways:

$$\dot{\sigma} = \sigma - l' \bigtriangledown_{\sigma} (L_{\text{Meta-E}}(f_{\dot{\phi}}(g_{\dot{\varphi}}(a_{\sigma}(T1, T2))), \text{target})) \quad (15)$$

where l' represents the learning rate of σ and $L_{\text{Meta-E}}$ is training $a_{\sigma}(\cdot)$ to encode the bitemporal image, defined as follows:

$$L_{\text{Meta-E}} = -\big[y \log(\varsigma(x)) + (1 - y) \log(1 - \varsigma(x))\big] \quad (16)$$

where ς(x) is the sigmoid function that converts logits to probability values: $\varsigma(x) = (1/(1 + \exp(-x)))$. y is the true label corresponding to $T1$ and $T2$ moments, and log denotes the natural logarithm. Overall, the goal of meta-learning is to enable the encoder Encode to learn the changing features, thereby improving the learning process of the encoder for the next iteration.

Adding a meta-learning algorithm to the network can better guide the model to solve the positive and negative sample imbalance problem, and experiments on its effectiveness are described in Section III-F3.

III. Experimental Results and Analysis

A. Dataset Acquisition

1) Public Datasets: To verify the effectiveness of the proposed method, we have chosen Google-CD [27], LEVIR-CD [28], and WHU-CD [29] datasets in our experiments, which are widely used recently.

1) *Google-CD:* This large-scale, multispectral, high-resolution dataset for change detection in satellite images has a resolution of 0.55 meters. It includes Guangzhou's suburbs, which saw a rapid increase in population from 2006 to 2019, mainly labeled as buildings that have undergone significant changes. The dataset collects 19 pairs of image pairs ranging in size from 1006 × 1168 to 4936 × 5224.
2) *WHU-CD:* This is a public RS BCD dataset created by Wuhan University. It consists of two large high-resolution (0.2 m/pixel) images taken in 2012 and 2016, with dimensions of 32 507 × 15 345 pixels. The dataset captures building changes in Christchurch, New Zealand, following a magnitude 6.3 earthquake in 2011 and subsequent extensive reconstruction efforts.
3) *LEVIR-CD:* The dataset contains 637 high-resolution (0.5 m) image pairs of 1024 × 1024. These RS images are from 20 different areas in various cities in Texas, USA, and cover the changes in various buildings from 2002 to 2018. Of these, 445 pairs were defined for training and the rest for testing.

TABLE I
Experimental Environment Settings

operating system	Ubuntu 18.04
hardware environment	Intel Core I9 processor 500G Storage NVIDIA RTX 3090 GPU 24G video memory
software environment	CUDA Toolkit 11.4 CUDNN V7.4.2 Python 3.7 Pytorch-GPU 1.12.1

We cropped the images to 256 × 256. This allowed us to process the images efficiently without exceeding the memory limits of our GPU. After processing, Google-CD contains 3053 pairs of images, which are randomly disrupted to 2305 pairs of images for training and 748 pairs of images for testing. The WHU-CD dataset contains 4478 pairs of images for training and 2956 pairs of images for testing. The LEVIR-CD contains 7120 pairs of images for training set and 2048 pairs of images for testing.

2) Self-Built Dataset: To verify the effectiveness of the proposed method in real BCD, we used Google Earth API [50], [51] to collect 40 high-resolution image pairs with the size of 1024 × 768 from Changsha in China in the past 10 years. Then, each map was cut into 12 images of 256 × 256 size on average using a segmentation algorithm, and finally 480 pairs of images were obtained.

B. Experimental Environment and Evaluation Metrics

1) Experimental Environment: The performance of Meta-SGNet was evaluated through a series of experiments conducted in a consistent hardware and software environment. The specific environment parameters are shown in Table I.

2) Evaluation Metrics: In this study, the primary evaluation metric is the $F1$-score for the change category. We also recognize the imbalance between changing and unchanging pixels; therefore, we introduce mean Intersection over Union (mIOU) as an additional evaluation metric. mIOU can offset the data imbalance effect, reflecting the model's ability to handle the BCD task more objectively. $F1$-score is calculated from the precision and recall, while mIOU indicates the mean intersection ratio. Additionally, we consider precision, recall, intersection, and union ratio of the change region (cIOU), and intersection and union ratio of the unchanged region (ucIOU) as important reference metrics. These are also regarded as critical criteria. The definitions of these metrics are outlined below

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \quad (17)$$

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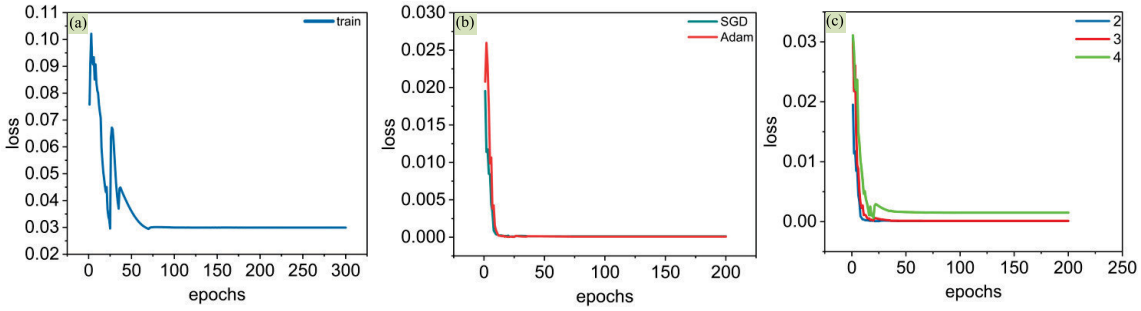


Fig. 3. (a) Selection of the number of training rounds. (b) Loss performance plot for the SGD and Adam optimizer networks. (c) Loss performance plot for the network when batch size is 2, 3, and 4.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{18}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{19}$$

$$F1\text{-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{20}$$

$$\text{cIOU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{21}$$

$$\text{ucIOU} = \frac{\text{TN}}{\text{FP} + \text{TN} + \text{FN}} \tag{22}$$

$$\text{mIOU} = \frac{\text{cIOU} + \text{ucIOU}}{2} \tag{23}$$

where true positive (TP) denotes the number of building pixels that were correctly identified as having changed, and false positive (FP) denotes the number of building pixels that were incorrectly considered to have changed. True negatives (TNs) and false negatives (FNs) denote the number of pixels correctly categorized as unchanged and incorrectly categorized as unchanged, respectively.

C. Parameter Settings

In this study, the choice of hyperparameters is crucial to the results. To select suitable experimental hyperparameters, we set the initial learning rate to 0.01, momentum to 0.9, and weight decay rate to 10^{-5}. The learning rate decayed at the 20th, 35th, 70th, and 95th iterations with a decay rate of 0.1. In terms of the number of training cycles, optimization method, and batch size settings, we have determined through the following experiments.

To determine the appropriate number of training cycles, the model is trained for 300 epochs. As shown in Fig. 3(a), the loss of the network basically stabilizes when the training reaches 200 cycles. Therefore, we decided to set the number of training cycles for the network to 200.

We improve the network parameters using the stochastic gradient descent method (SGD) and the adaptive moment estimation algorithm (Adam) to approximate the optimal value of the model output. As shown in Fig. 3(b), the network loss using both algorithms gradually stabilizes as the number of iterations increases. However, when Adam is used, the loss curve shows obvious oscillations and fluctuations. In contrast, the loss curve of the network trained with SGD is smoother and converges faster.

Regarding parameter updates, small batch training increases the frequency of parameter updates, accelerating network convergence. In contrast, large batch training provides more reliable gradient estimation. To determine the appropriate batch size, we conducted experiments with batch sizes of 2, 3, and 4. As shown in Fig. 3(c), the network losses are similar for batch sizes 2 and 3, but at a batch size of 4, the losses fluctuate more and are higher. Although the loss value at a batch size of 3 is higher with less fluctuation, we selected a batch size of 2 as the optimal solution.

D. Performance Experiments

In this section of experiments, the Google-CD dataset was randomly rearranged, and then Mete-SGNet and backbone were trained using a fivefold cross-validation method for 200 epochs at a time. The only difference of backbone compared to Mete-SGNet is the removal of SCCM, GFFM, and Meta-E. This process produced five sets of experimental data, as shown in Fig. 4, and the results demonstrate the good stability of Meta-SGNet. The average accuracy of these five training/testing sets was then calculated and the results are shown in Fig. 5(a) and (b). Observing the results, although backbone and Meta-SGNet models perform similarly in training accuracy, Meta-SGNet significantly outperforms backbone in terms of stability and validation accuracy. This indicates that the model training based on Meta-SGNet is more stable and efficient, with stronger generalization ability. Second, as shown in Fig. 5(b)–(d), compared to backbone, Meta-SGNet shows less fluctuation in accuracy, $F1$-score, and cIOU between the 50th and 200th epochs of the test. In contrast, the accuracy of Meta-SGNet outperforms backbone.

E. Compare With SOTA Methods

We compare the proposed method in this article with three primitive fully convolutional Siamese networks and ten other recent BCD methods, including FC-EF [52], fully convolutional Siam-difference (FC-Siam-D) [52], fully convolutional Siamese-concatenation (FC-Siam-C) [52], STA-Net [28], bitemporal image transformation (BIT) [53], HFA-Net [31], FERA-Net [34], HANet [40],GAS-Net [54], AMT-Net [55], AFCF3D-Net [32], MSDFFN [36], and DCENet [33].

We trained all models for 200 rounds and selected the best-performing model from the testing set for each method. To ensure fairness, given the significant impact of

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TABLE II
COMPARISON RESULTS ON GOOGLE-CD DATASET

Methods	Acc(%)	F1(%)	Re(%)	Pre(%)	cIOU(%)	UcIOU(%)	mIOU(%)
FC-EF (2018)	93.52	66.72	74.33	60.53	50.06	93.07	71.57
FC-Siam-D(2018)	94.26	66.93	66.43	67.43	50.29	93.91	72.10
FC-Siam-C(2018)	91.89	65.12	68.60	61.98	48.28	91.23	69.75
STA-Net (2020)	93.08	74.43	84.23	62.86	59.28	92.31	75.79
BIT (2021)	95.32	79.97	84.60	75.82	66.63	94.84	80.74
HFA-Net (2022)	96.03	82.93	83.26	79.01	70.84	95.61	83.23
FERA-Net (2023)	94.48	76.91	83.29	71.44	62.48	93.92	78.20
HANet (2023)	94.89	78.16	82.88	73.95	64.15	94.37	79.27
AMT-Net(2023)	96.45	83.93	83.96	83.90	72.31	96.09	84.20
GAS-Net(2023)	96.67	84.94	84.02	85.58	73.89	96.21	85.05
AFCF3DNet(2023)	96.51	84.88	81.20	88.91	73.73	96.12	84.93
MSDFFN(2023)	96.33	84.16	**88.39**	80.32	72.66	95.93	84.30
DCENet(2024)	96.64	85.24	82.83	87.81	74.28	96.28	85.28
Meta-SGNet	**97.01**	**86.84**	85.13	**89.22**	**75.63**	**96.70**	**86.17**

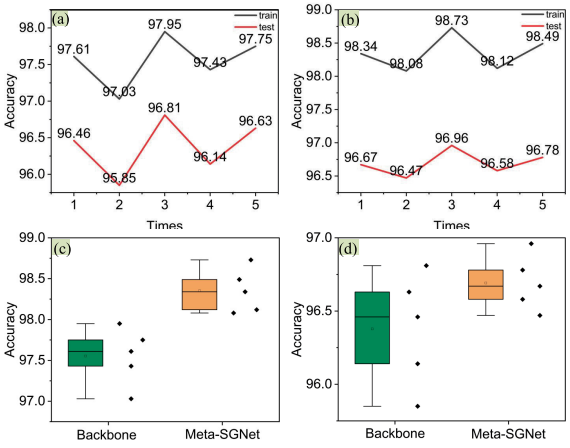


Fig. 4. Accuracy values of backbone and Meta-SGNet for fivefold cross-validation training and testing on Google-CD dataset. (a) Accuracy values for training. (b) Accuracy values for testing. (c) Accuracy box plots for training. (d) Accuracy box plots for testing.

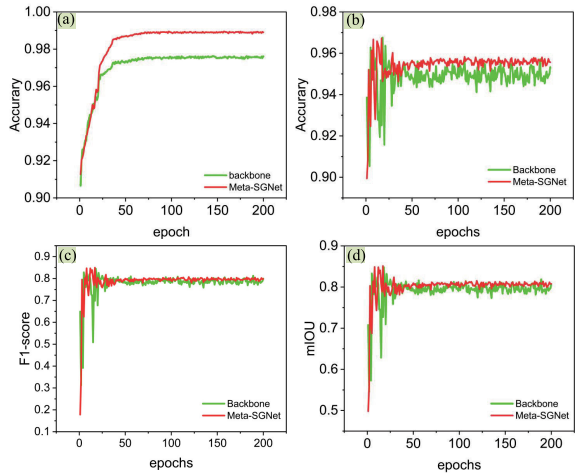


Fig. 5. (a) Curve of Meta-SGNet and backbone training 200 epochs on Google-CD dataset. (b)–(d) Accuracy, $F1$-score, and mIOU values tested by Meta-SGNet and backbone after training 200 epochs in Google-CD dataset.

hyperparameters on model performance, we used public code with default hyperparameters to implement the above CD networks.

1) Comparison Experiments on the Google-CD Dataset: Table II shows the results of the calculations of various methods on the Google-CD dataset. Our proposed method is superior to other methods in all the evaluation indicator, indicating that Meta-SGNet exhibits stronger robustness in the case of insufficient data. The accuracy of Meta-SGNet on the Google-CD reaches 97.01%, the $F1$-score reaches 87.12%, and mIOU reaches 86.17%.

Visualization results for each method are shown in Fig. 6. For clarity, we use white for changing buildings (TP), black for background (TN), red for false detection areas (FP), and green for missed detection areas (FN). Meta-SGNet achieves better results than other models in the Google-CD dataset. First, since the appearance of invariant objects is similar to the appearance of changing buildings, it is easy to have false alarms when performing change detection, which can be effectively avoided by our proposed model. For example, in Fig. 6(a) and (d), roads are mistaken for building changes (labeled in red) because roads and buildings have very similar morphological features. This similarity makes it difficult for some comparative methods to exclude roads, resulting in misclassification. Second, as shown in Fig. 6(a), (b), and (d), traditional methods often misidentify irrelevant changes, caused by seasonal differences or appearance changes, as building changes (labeled in red) because they cannot effectively learn contextual information. However, Meta-SGNet is able to learn contextual information well in the spatiotemporal domain, which can capture real semantic changes and filter out irrelevant changes. Finally, for a large range of changes, Meta-SGNet can generate complete prediction results, e.g., in Fig. 6(c), due to the limitation of model sensitivity, some comparison methods cannot fully detect the larger building area in the graph (labeled in green), while Meta-SGNet gives more complete results. In addition, while other methods can better distinguish the edges of the building from its surroundings, the Meta-SGNet recognition results have a clearer edge detection, which can be clearly seen in Fig. 6(c) and (d).

2) Comparison Experiments on WHU-CD and LEVIR-CD Datasets: We conducted comparison experiments on WHU-CD and LEVIR-CD datasets to validate the robustness of the proposed method. Table III shows the comparison results for the WHU-CD dataset and the LEVIR-CD dataset, respectively.

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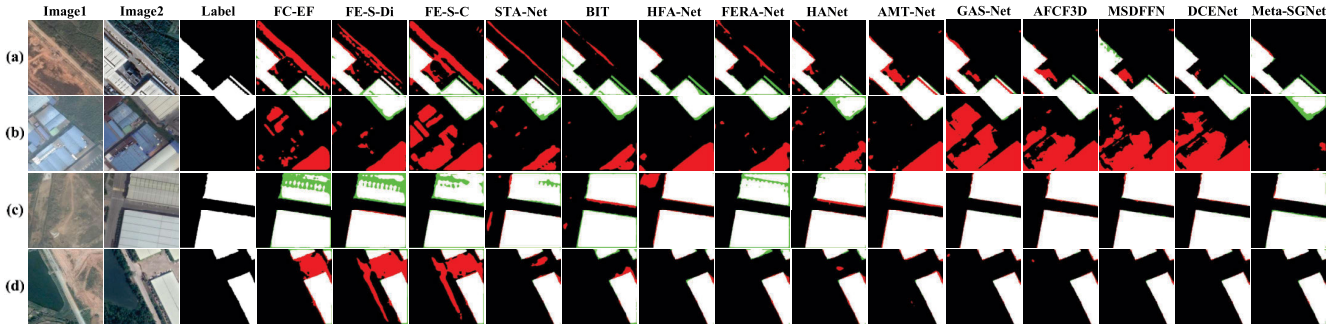


Fig. 6. Some of the experimental result plots of the Google-CD dataset on our Meta-SGNet network as well as on thirteen other networks, from left to right: the $T1$ moment image, the $T2$ moment image, and the labels FC-EF, FE-Siam-Di, FE-Siam-C, STA-Net, BIT, HFA-Net, FERA-Net, HANet, AMT-Net, GAS-Net, AFCF3D, MSDFFN, DCENet, and Meta-SGNet, where TP (white) denotes the changed buildings, TN (black) denotes the background, FP (red) denotes the wrongly detected areas, and FN (green) denotes the missed areas. (a)–(d) Represent four sets of visualizations selected from the experimental results, respectively.

TABLE III
COMPARISON RESULTS ON WHU-CD AND LEVIR-CD DATASETS

Methods	WHU-CD	LEVIR-CD
	Acc/F1/Re/Pre/cIOU/UcIOU/mIOU (%)	Acc/F1/Re/Pre/cIOU/UcIOU/mIOU (%)
FC-EF (2018)	97.42/70.80/83.89/61.24/54.80/97.33/76.07	98.60/85.11/81.08/89.71/74.08/98.54/86.31
FC-siam-D(2018)	94.76/56.03/89.57/40.76/38.92/94.58/66.75	98.41/84.50/84.64/84.35/73.16/98.34/85.75
FC-Siam-C(2018)	96.49/65.14/87.97/51.72/48.31/96.37/72.34	98.59/85.61/81.88/89.68/74.84/98.53/86.68
STA-Net (2020)	98.41/76.86/70.89/83.94/62.42/98.36/80.39	98.78/87.40/84.53/90.56/77.63/98.73/88.18
BIT (2021)	98.43/76.24/67.68/87.54/61.81/98.23/80.02	98.75/86.79/80.45/91.20/76.66/98.69/87.68
HFA-Net (2022)	98.40/78.08/76.47/79.76/64.04/98.35/81.20	98.77/87.15/80.91/91.42/77.23/98.73/87.97
FERA-Net (2023)	98.21/71.79/60.79/**87.64**/56.00/98.17/77.08	98.46/84.79/83.91/85.69/73.60/98.39/86.00
HANet (2023)	98.39/77.53/74.29/81.06/63.31/98.31/80.83	98.51/85.18/83.62/86.81/74.19/98.45/86.32
AMT-Net(2023)	98.21/78.38/86.77/71.46/64.44/98.15/81.30	98.79/87.53/83.26/91.86/77.82/98.73/88.28
GAS-Net(2023)	98.12/78.17/89.56/69.11/64.16/98.06/81.11	98.69/87.99/85.74/82.91/78.57/98.63/88.60
AFCF3DNet(2023)	98.05/77.69/**90.85**/67.86/63.52/97.98/80.75	98.56/86.92/**93.78**/81.01/76.87/98.49/87.68
MSDFFN(2023)	98.11/78.04/89.71/69.05/63.99/98.05/81.02	98.90/85.94/89.58/82.58/75.84/98.39/87.12
DCENet(2024)	98.20/78.57/88.46/70.68/64.71/98.14/81.42	98.70/87.71/91.03/84.63/78.12/98.63/88.38
Meta-SGNet	**98.46/79.68**/85.37/74.61/**66.21/98.41/82.31**	**98.93/89.20**/86.48/**92.02/80.50/98.90/89.70**

It can be seen that our method demonstrates excellent performance in both WHU-CD and LEVIR-CD and outperforms other methods in evaluation metrics such as mIOU and $F1$-score, and our accuracy score reaches 98.46%, $F1$-score 79.68%, and mIOU 82.31% on the WHU-CD dataset; on the LEVIR-CD dataset, our accuracy score reached 98.93%, $F1$-score reached 89.20%, and mIOU score reached 89.70%. Therefore, our proposed method has generality in BCD.

The visualized comparison results on the WHU-CD and LEVIR-CD datasets are shown in Fig. 7. To observe better comparison results, white, black, red, and green are also used to label the prediction results. It can be observed that Meta-SGNet's model achieves good results on both datasets as well. First, the model proposed in this article can better avoid FPs due to the similarity between the appearance of the object and the appearance of change. In Fig. 7(a) and (c), roads are mistaken for building changes (labeled in red) due to the fact that roads and buildings have very similar morphological features, which are difficult to be excluded by some comparative methods, resulting in misclassification of prediction results. Second, our Meta-SGNet can also deal well with irrelevant changes caused by seasonal differences or appearance changes as in Fig. 7(b) and (d)–(h). Then, for relatively small and intensive changes, our method also shows good results as shown in Fig. 7(g) and (h). Finally, for a wide range of variations, our Meta-SGNet can generate relatively complete predictions, as shown in Fig. 7(c) and (f). In addition, when dealing with edges in and around buildings, Meta-SGNet provides clearer edge detection compared to other methods, as shown in Fig. 7(a)–(c) and (f)–(h).

F. Method Effectiveness Experiments

To verify the effectiveness and advancement of the three methods, we conducted comparative experiments and validation of the effectiveness of each of the three methods on the Google-CD dataset.

1) Effectiveness Experiment of SCCM: In the validity experiments of the method, to verify the advancement and validity of our proposed SCCM, we replaced the convolution module of the encoder in the fully convolutional Siamese architecture with the SCCM, SA [31], HF [31], HFAB [31], and STA [28], and performed quantitative comparison experiments.

The results of the quantitative comparison experiments are shown in Table IV, where our proposed SCCM outperforms in all evaluation metrics. Among them, STA has a smaller impact on the results, only improving by about 0.3 percentage points; SA and HFAB improved by about 1.2 percentage points and HF improved by about 1.7 percentage points. The

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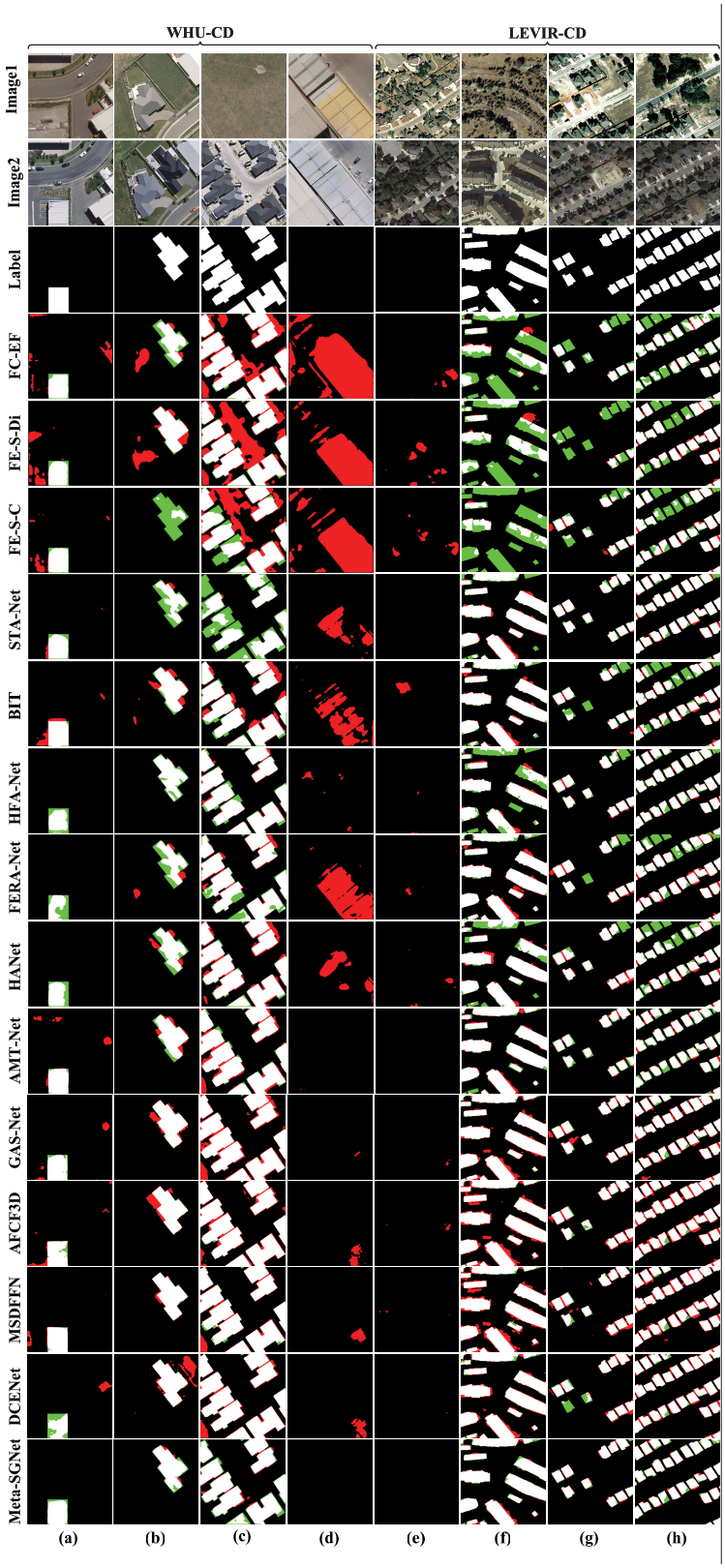


Fig. 7. Part of the experimental result images of WHU-CD and LEVIR-CD datasets on our Meta-SGNet network as well as on 13 other networks, from up to down: image at the moment of $T1$, image at the moment of $T2$, label, FC-EF, FE-Siam-Di, FE-Siam-C, STA-Net, BIT, HFA-Net, FERA-Net, HANet, AMT-Net, GAS-Net, AFCF3D, MSDFFN, DCENet, and Meta-SGNet, where TP (white) denotes the changed buildings, TN (black) denotes the background, FP (red) denotes the wrongly detected areas, and FN (green) denotes the missed areas. (a)–(d) Represent the four sets of visualizations selected from the WHU-CD experimental results and (e)–(h) represent the four visualizations selected from the LEVIR-CD experimental results, respectively.

improvement effect of adding SCCM to the full convolutional Siamese network is more obvious, in which Acc improves by 0.8 points and all other indicators improve by about 2 points. From Table IV, it can be concluded that SCCM is more capable of extracting features in high-resolution RS images with complex backgrounds.

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TABLE IV
RESULTS OF SCCM EFFECTIVENESS COMPARISON

Methods	Acc(%)	F1(%)	Re(%)	Pre(%)	cIOU(%)	UcIOU(%)	mIOU(%)
Backbone	95.96	77.92	77.62	78.33	70.53	95.53	83.03
Backbone+HF	96.59	84.58	**84.62**	84.53	73.28	96.24	84.76
Backbone+SA	96.53	83.86	81.59	86.27	72.21	96.19	84.21
Backbone+HFAB	96.51	83.94	82.65	85.27	72.33	96.16	84.24
Backbone+STA	96.23	82.87	82.67	83.07	70.76	95.85	83.30
Backbone+SCCM	**96.76**	**84.80**	83.83	**85.99**	**73.61**	**96.44**	**85.03**

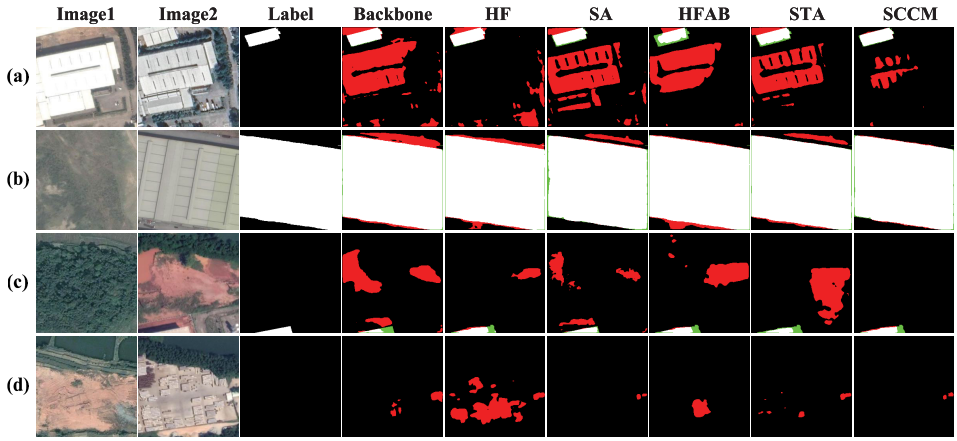


Fig. 8. Comparison of the effectiveness of SCCM with other modules in building feature extraction, from left to right: image taken at time $T1$, image taken at time $T2$, label, backbone, HF, SA, HFAB, STA, and SCCM, where TP (white) denotes the changed buildings, TN (black) denotes the background, FP (red) denotes the wrongly detected areas, and FN (green) denotes the missed areas. (a)–(d) Represent four sets of visualizations selected from the experimental results, respectively.

TABLE V
RESULTS OF GFFM EFFECTIVENESS COMPARISON

Methods	Acc(%)	F1(%)	Re(%)	Pre(%)	cIOU(%)	UcIOU(%)	mIOU(%)
Backbone	95.96	77.92	77.62	78.33	70.53	95.53	83.03
Backbone+Add	95.10	76.32	71.59	81.71	61.71	94.68	78.19
Backbone+Sub	96.23	83.21	84.68	81.79	71.25	95.84	83.55
Backbone+FEM	96.53	83.76	84.11	83.59	72.06	96.15	84.10
Backbone+Fusion	96.45	83.93	83.96	**83.90**	72.31	96.09	84.20
Backbone+GFFM	**96.52**	**84.08**	**84.92**	83.26	**73.98**	**96.18**	**85.08**

Some of the visualization results are shown in Fig. 8, where we marked the sections with different colors for better observation: changing buildings (TP, white), background (TN, black), misdetections (FP, red), and missed detections (FN, green). As shown in Fig. 8, the red area indicates the false detection area. Due to the interference of complex backgrounds, the appearance of some objects is similar to the actual change area, resulting in false detections in the model. In contrast, SCCM, through its unique structure, is able to more effectively distinguish between irrelevant and truly variable areas. SCCM can not only generate more discriminatory, but also effectively aggregate remote context information, which can help the model more accurately understand the changing characteristics of the region of interest. The results in Fig. 8 show that SCCM is more robust and accurate than other feature extraction modules when faced with the challenges of complex backgrounds and similar appearances. This proves the superiority of SCCM in BCD tasks.

2) Effectiveness Experiment of GFFM: To validate our proposed GFFM and effectiveness, we replace concatenation in the fully convolutional Siamese architecture with addition(Add) [52], subtraction(Sub) [52], FEM [34], Fusion modules [37], and GFFM, and perform comparison experiments.

Experimental results in Table V show that Sub, FEM, and Fusion can improve the metrics, but with GFFM the effect is higher than these modules. Add is not suitable for feature fusion operations in fully convolutional concatenated networks. The metrics show that GFFM retains task-relevant feature properties better than these fusion modules.

Some of the visualization results are shown in Fig. 9, where we marked the sections with different colors for better observation: changing buildings (TP, white), background (TN, black), misdetections (FP, red), and missed detections (FN, green). As shown in Fig. 9, the green areas represent the undetected areas. Due to the information redundancy in the process of dual-phase feature fusion, some traditional

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TABLE VI
RESULTS OF META-E EFFECTIVENESS COMPARISON

Methods	Acc(%)	F1(%)	Re(%)	Pre(%)	cIOU(%)	UcIOU(%)	mIOU(%)
Backbone	95.96	77.92	77.62	78.33	70.53	95.53	83.03
Backbone+GAM	96.48	83.09	78.30	88.11	71.08	96.15	83.61
Backbone+FAM	96.30	84.13	80.84	79.89	72.61	95.90	84.25
Backbone+PFBD	96.31	82.92	81.16	84.75	70.82	95.95	83.38
Backbone+Meta-E	**96.80**	**84.96**	**81.77**	**88.42**	**73.86**	**96.49**	**85.18**

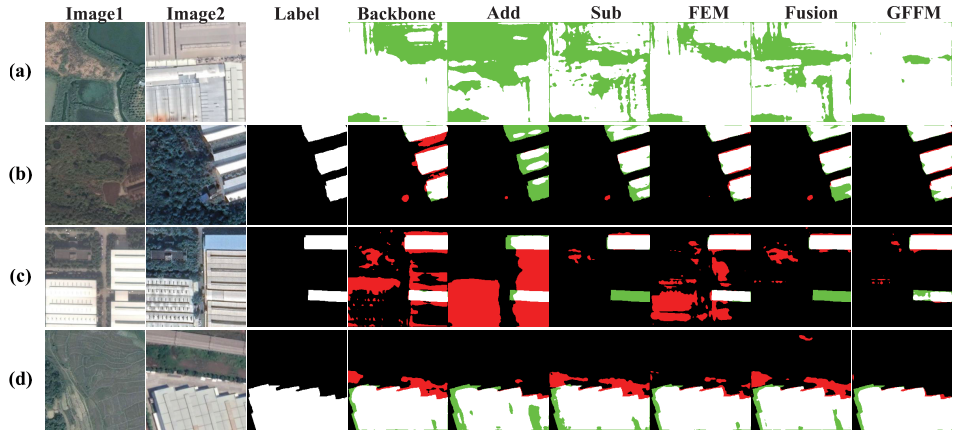


Fig. 9. Comparison of the effectiveness of GFFM with other modules in building feature extraction, from left to right: image at time $T1$, image at time $T2$, label, backbone, Add, Sub, FEM, Fusion, and GFFM, where TP (white) denotes the changed buildings, TN (black) denotes the background, FP (red) denotes the wrongly detected areas, and FN (green) denotes the missed areas. (a)–(d) Represent four sets of visualizations selected from the experimental results, respectively.

models cannot effectively distinguish and take advantage of the important differences of different phase features, which leads to the redundancy of features and the phenomenon of missed detection in the detection process. In contrast, GFFM performs dynamic feature selection through gated values, which effectively avoids the problem of information redundancy and ensures that all key features are fully considered. GFFM not only dynamically adjusts feature weights from different time points, but also retains more details during the fusion process, which makes it particularly good at dealing with complex bitemporal feature fusion problems. The comparison results in Fig. 9 show that GFFM has significant advantages over other feature fusion modules in the BCD task, which can significantly reduce missed detections and improve the comprehensiveness and accuracy of detection.

3) Effectiveness Experiment of Meta-E: To validate the effectiveness and advantages of our proposed Meta-E, we compare it with several methods that have been used for the sample imbalance, including GAM [54], foreground awareness module (FAM) [54], and PFBS [40]. These algorithms are added to the Backbone network and quantitative comparison experiments are conducted.

The experimental results in Table VI show that although the GAM, FAM, and PFBS modules can all improve the detection indicators to a certain extent, the improvement of these modules is relatively small compared with Meta-E. Specifically, Meta-E is able to adaptively adjust the model's attention to positive samples through its unique meta-learning encoder design, so as to better deal with the sample imbalance problem. This self-learning mechanism allows Meta-E to significantly improve the accuracy and robustness of the detection when dealing with sample imbalances. Experimental data shows that Meta-E shows obvious advantages in dealing with sample imbalance problems, proving its superiority in BCD tasks.

4) Ablation Experiments: We tested the proposed module on Google-CD to confirm its effectiveness. Google-CD is relatively small but includes a diverse range of building types, posing additional challenges to our approach. Since our proposed SCCM and Meta-E modules did not alter the input data dimensions, we easily adapted the network structure for ablation experiments. We used the same hyperparameters to train multiple backbone networks with different modifications on the Google-CD dataset for 200 generations and selected the best-performing model on the testing set. Table VII shows the experimental results.

The experimental results show that SCCM, GFFM, and Meta-E in Meta-SGNet positively affect the detection ability of the model. As shown in Table VII, the detection results of the Backbone network perform well, but the accuracy, $F1$-score, and mIOU are lower than those of adding the proposed modules. This indicates a limitation in the performance of the backbone network in the BCD tasks. The addition of SCCM improved all evaluation metrics (accuracy: +0.80%, $F1$-score: +2.08%, mIOU: +2.00%). It is clear that the addition of SCCM helps to improve the performance of the network. Adding GFFM to the backbone network will also improve the metrics (accuracy: +0.56%, $F1$-score: +1.36%, mIOU: +2.05%). This shows that the fusion of GFFM removes task-irrelevant features and pays more attention to task-relevant features. The metrics also improved when the images were encoded using the meta-learning guidance encoder (accuracy: +0.84%,

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TABLE VII
ABLATION EXPERIMENTS

Methods	Acc(%)	F1(%)	Re(%)	Pre(%)	cIOU(%)	UcIOU(%)	mIOU(%)
Backbone	95.96	77.92	77.62	78.33	70.53	95.53	83.03
Backbone+SCCM	96.76	84.80	83.83	85.99	73.61	96.44	85.03
Backbone+GFFM	96.52	84.08	84.92	83.26	73.98	96.18	85.08
Backbone+Meta-E	96.80	84.96	81.77	88.42	73.86	96.49	85.18
Backbone+SCCM +GFFM	96.83	85.62	85.05	85.70	74.86	96.50	85.68
Backbone+SCCM+Meta-E	96.91	85.53	82.63	88.64	74.72	96.60	85.66
Backbone+GFFM+Meta-E	96.86	85.54	83.97	87.15	74.73	96.54	85.64
Meta-SGNet	**97.01**	**86.12**	**85.13**	**89.22**	**75.63**	**96.70**	**86.17**

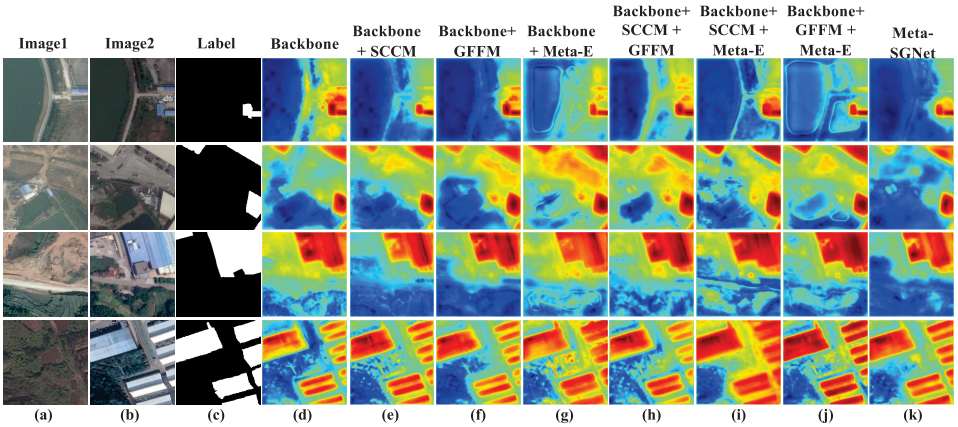


Fig. 10. (a)–(k) Represent the visualization of the ablation experimental results, in order: T1 moment image, T2 moment image, label, Backbone, Backbone+SCCM, Backbone+GFFM, Backbone+Meta-E, Backbone+SCCM+GFFM, Backbone+SCCM+Meta-E, Backbone+GFFM+Meta-E, MetaSGNet.

$F1$-score: +2.24%, mIOU: +2.15%). Meanwhile, Meta-SGNet obtained the best accuracy, $F1$-score, and mIOU (accuracy: + 1.05%, $F1$-score: +3.40%, mIOU: +3.14%). These results further validate that the modules proposed in this article play a positive role in handling the BCD tasks.

Fig. 10 shows some of the visualization result maps of the ablation experiments on the Google-CD dataset, where Fig. 10(d) shows the result of the backbone network recognition. Compared with Fig. 10(d), the misdetection rate is significantly reduced with the addition of SCCM in Fig. 10(e), in addition to the regional characteristics of buildings that are more obvious. As shown in Fig. 10(f), compared with Fig. 10(d) of the backbone network, the misdetection rate is further reduced with the addition of GFFM, which eliminates more task-irrelevant features and also focuses more on task-relevant features. Comparing Fig. 10(d) with Fig. 10(g), it is clear that with the addition of Meta-E, the model focuses more on changing features. The local features of the buildings are more prominent, and the edges of the buildings are clearer compared to the irrelevant background. Finally, as shown in Fig. 10(k), the Meta-SGNet network largely improves the precision and accuracy of detection. Therefore, our proposed Meta-SGNet network is very effective.

G. Practical Application Analysis

After training and testing of Meta-SGNet, as shown in Fig. 11, we performed real BCD experiments using a self-built dataset. We input the segmented map images to the saved model (the model with the best results of Meta-SGNet tested on the Google-CD dataset) to get the output results.

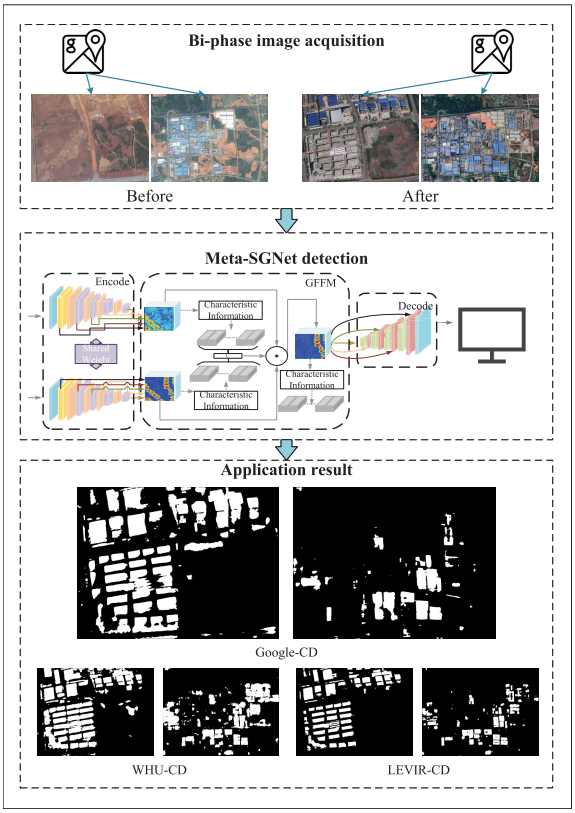


Fig. 11. Practical application results.

Then the change detection results were obtained by merging the algorithm to revert the 12 resultant images into 1024 × 768 size images. The quantitative results show that, on the

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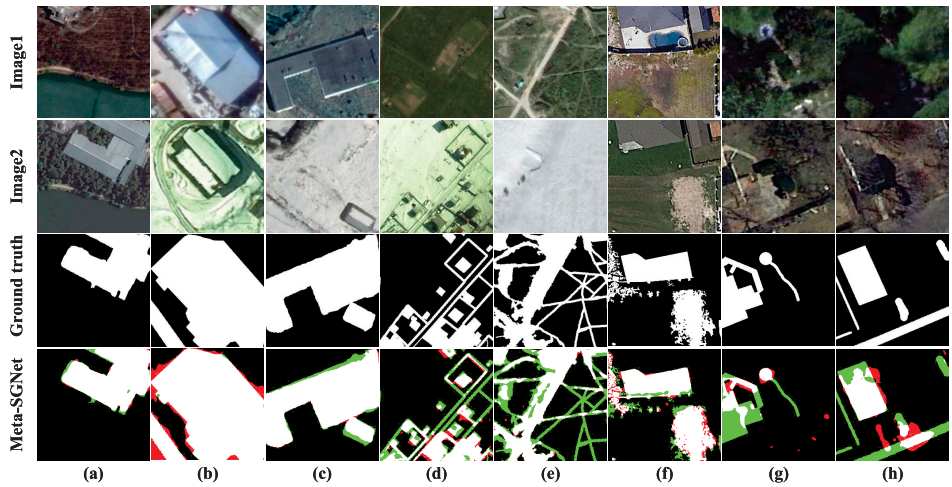


Fig. 12. Partial plot of the experimental results of the CDD dataset on the Meta-SGNet network, where TP (white) denotes the changed buildings, TN (black) denotes the background, FP (red) denotes the wrongly detected areas, and FN (green) denotes the missed areas. (a)–(h) Represent eight sets of visualizations selected from the experimental results, respectively.

model pretrained using the Google-CD dataset, the mIoU is 73.44%; on the model pretrained using the WHU-CD dataset, the mIoU is 69.04%; and on the model pretrained using the LEVIR-CD dataset, the mIoU is 77.03%. In Fig. 11, "Before" indicates the map before the change and "After" indicates the map after the change. The results show that Meta-SGNet can accurately capture the extent of building changes. Our detection algorithm still performs well even when there is complex background interference, lighting changes, and other factors in the change area. As shown in Fig. 11, in further experiments, we gradually expand the observation height to test the performance of the method more rigorously. During the process, we are able to capture building changes. Even with complex backgrounds and changing environments, the algorithm remains sensitive to changes, successfully achieving the goal of change detection. Overall, our proposed BCD algorithm demonstrates high adaptability and accuracy across different scenarios.

H. Discussion and Shortcomings

The Meta-SGNet proposed in this article is also applicable to other RS images in BCD tasks. We conducted additional experiments in this area using the CDD-CD dataset [56]. The dataset consists of multisource remotely sensed images, including 11 pairs of raw images, seven pairs of seasonal change images with dimensions of 4725 × 2200 pixels, and 4 pairs of images with dimensions of 1900 × 1000. The resolution varied between 3 centimeters and 100 centimeters per pixel, and the dual time-phase images varied considerably over the seasons. The CDD-CD dataset authors processed the raw data to generate a 256 × 256 dataset with a training set size of 10 000 and a testing set size of 3000 by trimming and rotating.

In the same experimental environment, we used Meta-SGNet to train on this dataset, and after 200 epochs of training, the generated image size is 256 × 256, as shown in Fig. 12. Experimental results show that Meta-SGNet performs well on the CDD-CD dataset, successfully detecting changes in dual-temporal images across seasons, as shown in Fig. 12(a)–(e). However, the performance of Meta-SGNet is relatively poor when dealing with images with complex backgrounds, road, grass, and forest changes, as shown in Fig. 12(d), (e), (g), and (h). Therefore, Meta-SGNet has a lack of fitting ability in coping with complex background images, and the generated images may appear incomplete and blurred. Future research will focus on background complexity processing to improve the generalization ability of the network, reduce the consumption of computational resources, and generate higher resolution images.

IV. Conclusion

In this article, a Meta-SGNet based on meta-learning is proposed to solve these problems in BCD, such as complex background, information redundancy in the fusion of bitemporal features, and data sample imbalance. The network structure has achieved excellent results in BCD. The summary is as follows.

1) The ablation experiments show that the SCCM can effectively distinguish between buildings and surface cover with similar spectral and morphological logic characteristics (accuracy: +0.80%, $F1$-score: + 2.08%, mIOU: +2.00%); the GFFM can avoid the data redundancy problem when fusing the features and the model has better recognition effect (accuracy: +0.56%, $F1$-score: +1.36%, mIOU: +2.05%); and the Meta-E module guides the encoder to learn more building features autonomously, which improves the accuracy of building detection (accuracy: +0.84%, $F1$-score: +2.24%, mIOU: +2.15%).
2) Compared with FC-Siam-C, our recognition accuracy is higher (accuracy: +5.12%, $F1$-score: + 21.00%, mIOU: +16.42%); compared with HFA-Net, our method is more advantageous for the recognition of ground cover with similar optical properties as buildings (accuracy: +0.98%, $F1$-score: +3.19%, mIOU: +2.94%); compared with FERA-Net, our method can distinguish the building boundaries well and has a lower miss and false detection rates (accuracy: +2.53%,

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$F1$-score: +9.21%, mIOU: +7.97%); and compared to GAS-Net, our method can better deal with positive and negative sample imbalance problems (accuracy: +0.34%, $F1$-score: +0.54%, mIOU: +1.12%). The above experimental results show that our network performs well in realistic scenarios.

3) Compared with the current popular BCD method, the average accuracy of our method on Google-CD, WHU-CD, and LEVIR-CD datasets is 97.01%, 98.46%, and 98.93%, $F1$-score is 86.12%, 79.68%, and 89.20%, and mIOU averages are 86.17%, 82.31%, and 89.70%, respectively.
4) The application results show that the method can accurately capture the range of building changes, and also shows a high adaptability and the ability to quickly detect building changes in different scenarios.

This article's method skillfully addresses three difficulties in BCD, but two key points deserve further discussion in future research. First, current training data for BCD often requires extensive labeling, which is costly, especially in large areas. Semisupervised and unsupervised methods can address this issue by using unlabeled data, reducing the model's dependence on labeled data and lowering modeling costs. Second, BCD results are usually generated automatically by the model, but users need to understand how these results are derived. Semantic understanding and interpretive techniques can help explain the model's decisions more comprehensibly so that users can understand the basis of the model's judgment. These two directions will provide new breakthrough points for future RS image change detection research.

REFERENCES

[1] G. Vivekananda, R. Swathi, and A. Sujith, "Multi-temporal image analysis for LULC classification and change detection," *Eur. J. Remote Sens.*, vol. 54, no. 2, pp. 189–199, Mar. 2021.

[2] Z. Chen et al., "EGDE-Net: A building change detection method for high-resolution remote sensing imagery based on edge guidance and differential enhancement," *ISPRS J. Photogramm. Remote Sens.*, vol. 191, pp. 203–222, Sep. 2022.

[3] Z. Li, H. Shen, Q. Cheng, Y. Liu, S. You, and Z. He, "Deep learning based cloud detection for medium and high resolution remote sensing images of different sensors," *ISPRS J. Photogramm. Remote Sens.*, vol. 150, pp. 197–212, Apr. 2019.

[4] R. Liu, M. Kuffer, and C. Persello, "The temporal dynamics of slums employing a CNN-based change detection approach," *Remote Sens.*, vol. 11, no. 23, p. 2844, Nov. 2019.

[5] X. Huang, Y. Cao, and J. Li, "An automatic change detection method for monitoring newly constructed building areas using time-series multi-view high-resolution optical satellite images," *Remote Sens. Environ.*, vol. 244, Jul. 2020, Art. no. 111802.

[6] Q. Zhu, C. Liao, H. Hu, X. Mei, and H. Li, "MAP-Net: Multiple attending path neural network for building footprint extraction from remote sensed imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 6169–6181, Jul. 2020.

[7] F. Xue, L. Wu, and W. Lu, "Semantic enrichment of building and city information models: A ten-year review," *Adv. Eng. Informat.*, vol. 47, Jan. 2021, Art. no. 101245.

[8] H. West, N. Quinn, and M. Horswell, "Remote sensing for drought monitoring & impact assessment: Progress, past challenges and future opportunities," *Remote Sens. Environ.*, vol. 232, Oct. 2019, Art. no. 111291.

[9] L. Rokach, O. Maimon, and E. Shmueli, *Machine Learning for Data Science Handbook: Data Mining and Knowledge Discovery Handbook*. Cham, Switzerland: Springer, 2023.

[10] S. K. Tasoulis, C. N. Doukas, V. P. Plagianakos, and I. Maglogiannis, "Statistical data mining of streaming motion data for activity and fall recognition in assistive environments," *Neurocomputing*, vol. 107, pp. 87–96, May 2013.

[11] G. Zhang, F. Tan, and Y. Wu, "Ship motion attitude prediction based on an adaptive dynamic particle swarm optimization algorithm and bidirectional LSTM neural network," *IEEE Access*, vol. 8, pp. 90087–90098, 2020.

[12] J. Li, Q. Li, Y. Zhu, Y. Ma, Y. Xu, and C. Xie, "An automatic extraction method of coach operation information from historical trajectory data," *J. Adv. Transp.*, vol. 2019, pp. 1–15, Feb. 2019.

[13] H. Xie and H. Huang, "Classification of land cover remote-sensing images based on pattern recognition," *Sci. Program.*, vol. 2022, pp. 1–15, Sep. 2022.

[14] Y. Wang, B. Xiao, A. Bouferguene, M. Al-Hussein, and H. Li, "Vision-based method for semantic information extraction in construction by integrating deep learning object detection and image captioning," *Adv. Eng. Informat.*, vol. 53, Aug. 2022, Art. no. 101699.

[15] L. Su, M. Gong, P. Zhang, M. Zhang, J. Liu, and H. Yang, "Deep learning and mapping based ternary change detection for information unbalanced images," *Pattern Recognit.*, vol. 66, pp. 213–228, Jun. 2017.

[16] J. Geng, H. Wang, J. Fan, and X. Ma, "Change detection of SAR images based on supervised contractive autoencoders and fuzzy clustering," in *Proc. Int. Workshop Remote Sens. Intell. Process. (RSIP)*, Shanghai, China, May 2017, pp. 1–3.

[17] M. Gong, T. Zhan, P. Zhang, and Q. Miao, "Superpixel-based difference representation learning for change detection in multispectral remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 5, pp. 2658–2673, May 2017.

[18] B. Hou, Y. Wang, and Q. Liu, "Change detection based on deep features and low rank," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 12, pp. 2418–2422, Dec. 2017.

[19] R. Liu, Z. Cheng, L. Zhang, and J. Li, "Remote sensing image change detection based on information transmission and attention mechanism," *IEEE Access*, vol. 7, pp. 156349–156359, 2019.

[20] X. Peng, R. Zhong, Z. Li, and Q. Li, "Optical remote sensing image change detection based on attention mechanism and image difference," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7296–7307, Nov. 2020.

[21] S. Li and L. Huo, "Remote sensing image change detection based on fully convolutional network with pyramid attention," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2021, pp. 4352–4355.

[22] Y. Hu et al., "Fast forest fire smoke detection using MVMNet," *Knowl.-Based Syst.*, vol. 241, Apr. 2022, Art. no. 108219.

[23] J. Deng, W. Huang, G. Zhou, Y. Hu, L. Li, and Y. Wang, "Identification of banana leaf disease based on KVA and GR-ARNet," *J. Integrative Agricult.*, vol. 23, no. 10, pp. 3554–3575, Oct. 2024.

[24] X. Pan, B. Zhong, Y. Wang, and L. Shen, "Identification of accident-injury type and bodypart factors from construction accident reports: A graph-based deep learning framework," *Adv. Eng. Informat.*, vol. 54, Oct. 2022, Art. no. 101752.

[25] M. Liu, X. Luo, G. Wang, and W.-Z. Lu, "Intelligent information extraction from government on-site inspection reports of construction projects: A graph-based text mining approach," *Adv. Eng. Informat.*, vol. 58, Oct. 2023, Art. no. 102163.

[26] H. Jiang et al., "A survey on deep learning-based change detection from high-resolution remote sensing images," *Remote Sens.*, vol. 14, no. 7, p. 1552, Mar. 2022.

[27] D. Peng, L. Bruzzone, Y. Zhang, H. Guan, H. Ding, and X. Huang, "SemiCDNet: A semisupervised convolutional neural network for change detection in high resolution remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5891–5906, Jul. 2021.

[28] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, 2020.

[29] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2018.

[30] X. He, Z. Tang, Y. Deng, G. Zhou, Y. Wang, and L. Li, "UAV-based road crack object-detection algorithm," *Autom. Construct.*, vol. 154, Oct. 2023, Art. no. 105014.

[31] H. Zheng et al., "HFA-net: High frequency attention Siamese network for building change detection in VHR remote sensing images," *Pattern Recognit.*, vol. 129, Sep. 2022, Art. no. 108717.

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[32] Y. Ye, M. Wang, L. Zhou, G. Lei, J. Fan, and Y. Qin, "Adjacent-level feature cross-fusion with 3-D CNN for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5618214.

[33] F. Luo, T. Zhou, J. Liu, T. Guo, X. Gong, and X. Gao, "DCENet: Diff-feature contrast enhancement network for semi-supervised hyperspectral change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5511514.

[34] X. Xu, Y. Zhou, X. Lu, and Z. Chen, "FERA-Net: A building change detection method for high-resolution remote sensing imagery based on residual attention and high-frequency features," *Remote Sens.*, vol. 15, no. 2, p. 395, Jan. 2023.

[35] M. Wang, B. Zhu, J. Zhang, J. Fan, and Y. Ye, "A lightweight change detection network based on feature interleaved fusion and bistage decoding," *IEEE J. Sel. Topics Appl. Earth Observat. Remote Sens.*, vol. 17, pp. 2557–2569, 2024.

[36] F. Luo, T. Zhou, J. Liu, T. Guo, X. Gong, and J. Ren, "Multiscale diff-changed feature fusion network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5502713.

[37] C. Zhang, L. Wang, S. Cheng, and Y. Li, "SwinSUNet: Pure transformer network for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5224713.

[38] H. Chen, W. Li, and Z. Shi, "Adversarial instance augmentation for building change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2021.

[39] H. Chen, W. Li, S. Chen, and Z. Shi, "Semantic-aware dense representation learning for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5630018.

[40] C. Han, C. Wu, H. Guo, M. Hu, and H. Chen, "HANet: A hierarchical attention network for change detection with bitemporal very-high-resolution remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observat. Remote Sens.*, vol. 16, pp. 3867–3878, 2023.

[41] A. Oubara, F. Wu, R. Maleki, B. Ma, A. Amamra, and G. Yang, "Enhancing adversarial learning-based change detection in imbalanced datasets using artificial image generation and attention mechanism," *ISPRS Int. J. Geo-Inf.*, vol. 13, no. 4, p. 125, Apr. 2024.

[42] Z. Li, K. Kamnitsas, and B. Glocker, "Overfitting of neural nets under class imbalance: Analysis and improvements for segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, Shenzhen, China. Cham, Switzerland: Springer, 2019, pp. 402–410.

[43] N. Ibtehaz and M. S. Rahman, "MultiResUNet: Rethinking the U-Net architecture for multimodal biomedical image segmentation," *Neural Netw.*, vol. 121, pp. 74–87, Jan. 2020.

[44] Q. Wang et al., "Segmentation network of concrete cracks with multi-frequency OctaveRes dual encoder and cross-attention mechanism optimized by average weight," *Autom. Construction*, vol. 155, Nov. 2023, Art. no. 105050.

[45] J.-J. Liu, Q. Hou, M.-M. Cheng, C. Wang, and J. Feng, "Improving convolutional networks with self-calibrated convolutions," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10096–10105.

[46] L. Zhang, C. Lu, H. Xu, A. Chen, L. Li, and G. Zhou, "MMFNet: Forest fire smoke detection using multiscale convergence coordinated pyramid network with mixed attention and fast-robust NMS," *IEEE Internet Things J.*, vol. 10, no. 20, pp. 18168–18180, Oct. 2023.

[47] X. Wei, W. Li, M. Zhang, and Q. Li, "Medical hyperspectral image classification based on end-to-end fusion deep neural network," *IEEE Trans. Instrum. Meas.*, vol. 68, no. 11, pp. 4481–4492, Nov. 2019.

[48] X. Ou, L. Liu, B. Tu, G. Zhang, and Z. Xu, "A CNN framework with slow-fast band selection and feature fusion grouping for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5524716.

[49] H. Gao, J. Li, W. Qiang, L. Si, F. Sun, and C. Zheng, "Bootstrapping informative graph augmentation via a meta learning approach," 2022, *arXiv:2201.03812*.

[50] J. Duque, J. Patino, and A. Betancourt, "Exploring the potential of machine learning for automatic slum identification from VHR imagery," *Remote Sens.*, vol. 9, no. 9, p. 895, Aug. 2017.

[51] G.-S. Xia et al., "DOTA: A large-scale dataset for object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3974–3983.

[52] R. C. Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.

[53] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2021.

[54] R. Zhang, H. Zhang, X. Ning, X. Huang, J. Wang, and W. Cui, "Global-aware Siamese network for change detection on remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 199, pp. 61–72, May 2023.

[55] W. Liu, Y. Lin, W. Liu, Y. Yu, and J. Li, "An attention-based multiscale transformer network for remote sensing image change detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 202, pp. 599–609, Aug. 2023.

[56] M. A. Lebedev, Y. V. Vizilter, O. V. Vygolov, V. A. Knyaz, and A. Y. Rubis, "Change detection in remote sensing images using conditional adversarial networks," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 42, pp. 565–571, May 2018.



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